Neuroinformatics Project 2025

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Outline

1. Data Visualization
2. Data Transformation (Neural Coding)
3. Transformed Data – Exploratory Analysis
4. Model Creation and Training
5. Model Evaluation
6. Model Comparision

Step 1:
Data Visualization

Plotting dataset x.npy

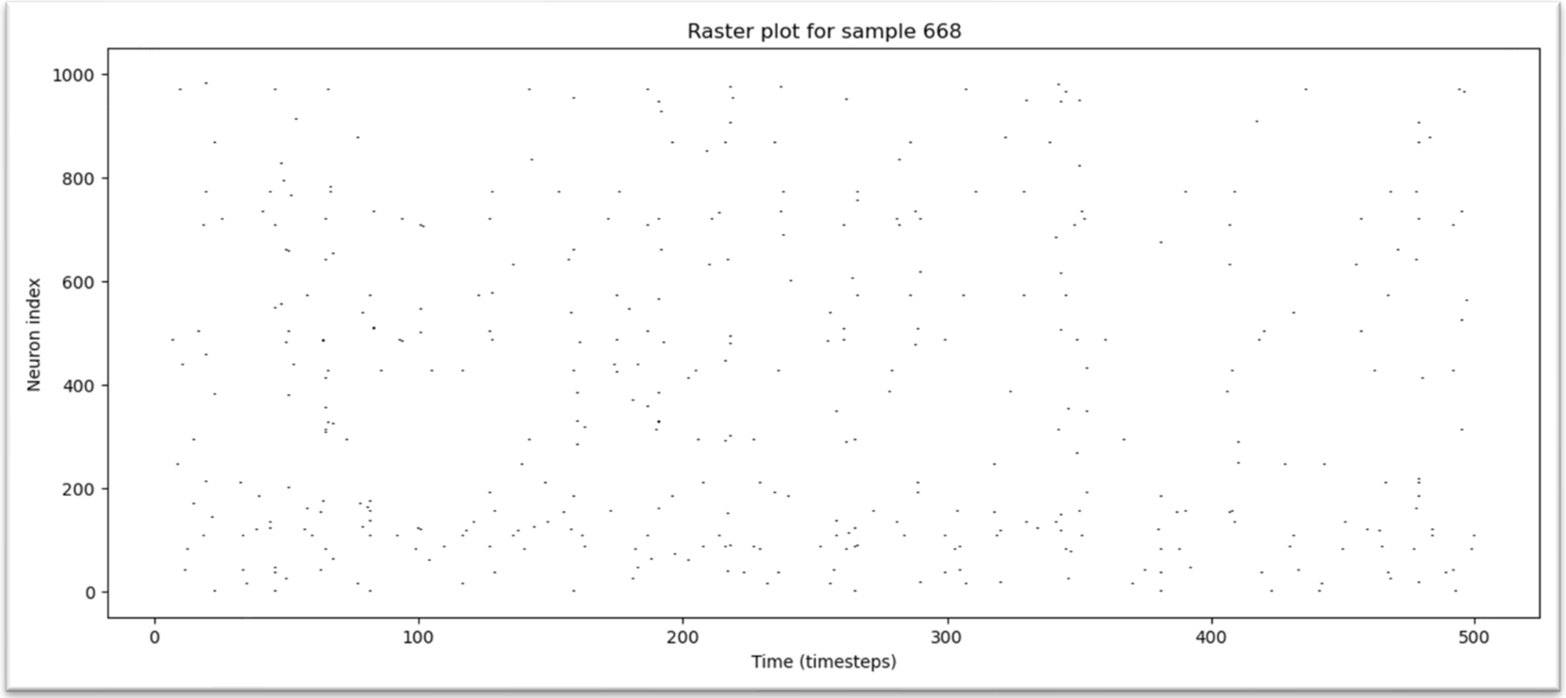
dataset contains neural firing activity from multiple neuraons in response to different stimuli (familiar or unfamiliar

shape: (800, 1000, 501)

800 experiment traisl

1000 neurons

501 timestamps



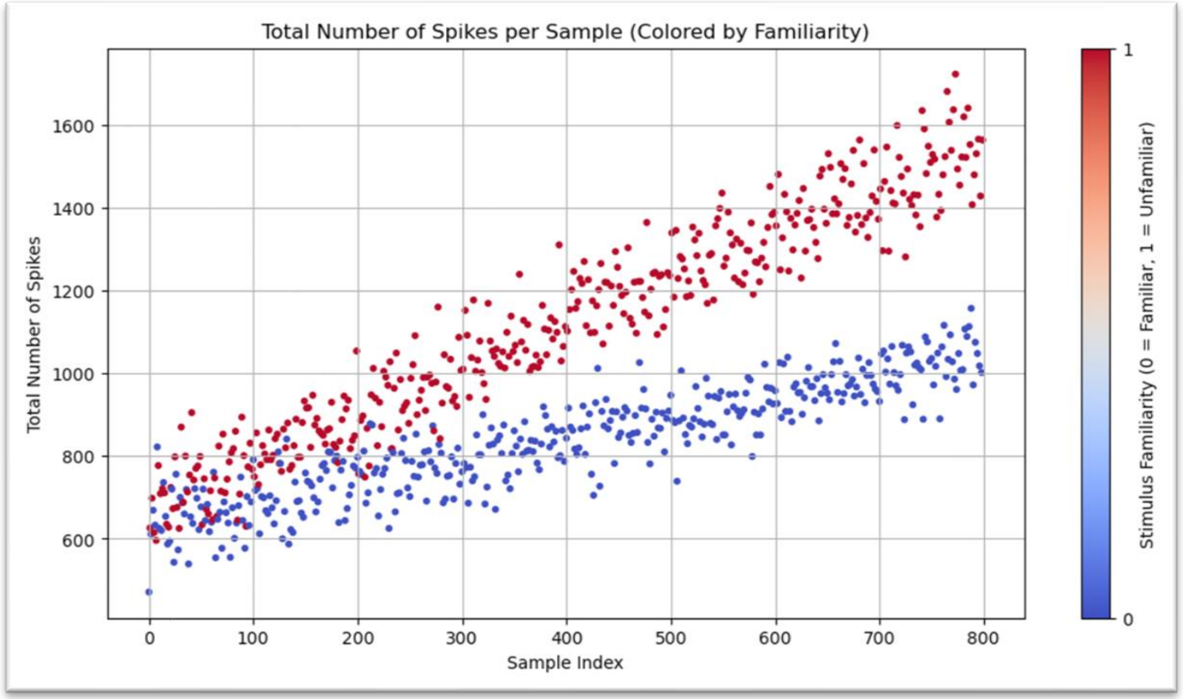
- raster plot visualizes the firing activity of all neurons during a randomly selected single experiment trial
- each vertical line represents a spike, showing the time at which a particular neuron fired

Plotting dataset x.npy

dataset contains neural firing activity from multiple neuraons in response to different stimuli (familiar or unfamiliar

shape: (800, 1000, 501)

- 800 experiment traisl
- 1000 neurons
- 501 timestamps



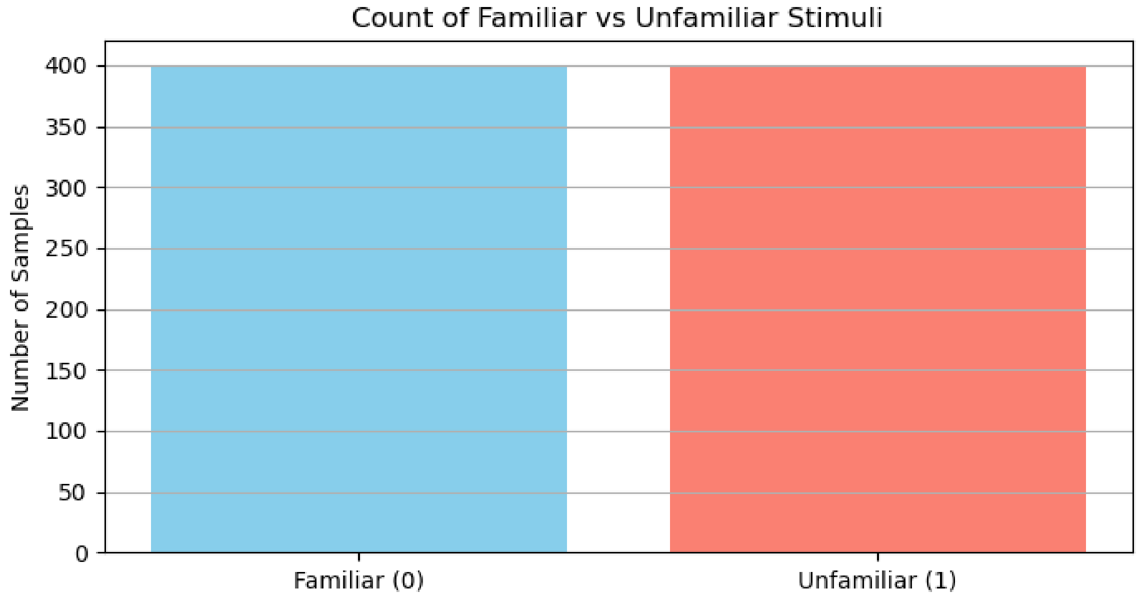
- total number of spikes recorded across all neurons and time points for each of the 800 experimental samples and colored according to stimulus familiarity
- plot reveals a clear separation between the two classes, with unfamiliar stimuli mostly producing a higher total number of spikes compared to familiar stimuli

Plotting dataset y.npy

dataset contains the labels corresponding to the stimuli presented in each experiment

shape: (800,)

each entry indicates whether the stimulus was familiar (label 0) or unfamiliar (label 1) to the neurons



- bar plot shows that an equal number of familiar and unfamiliar stimuli were used in the 800 experiments

Step 2: Data Transformation (Neural Coding)

Neural Coding

- refers to how the brain represents information through the activity of neurons (spikes)
- spikes can be encoded in different ways e.g. rate coding, temporal coding and burst coding

- I applied two different neural coding methods to the dataset x.npy to create datasets that can be used as the basis for a classifier

Rate Coding

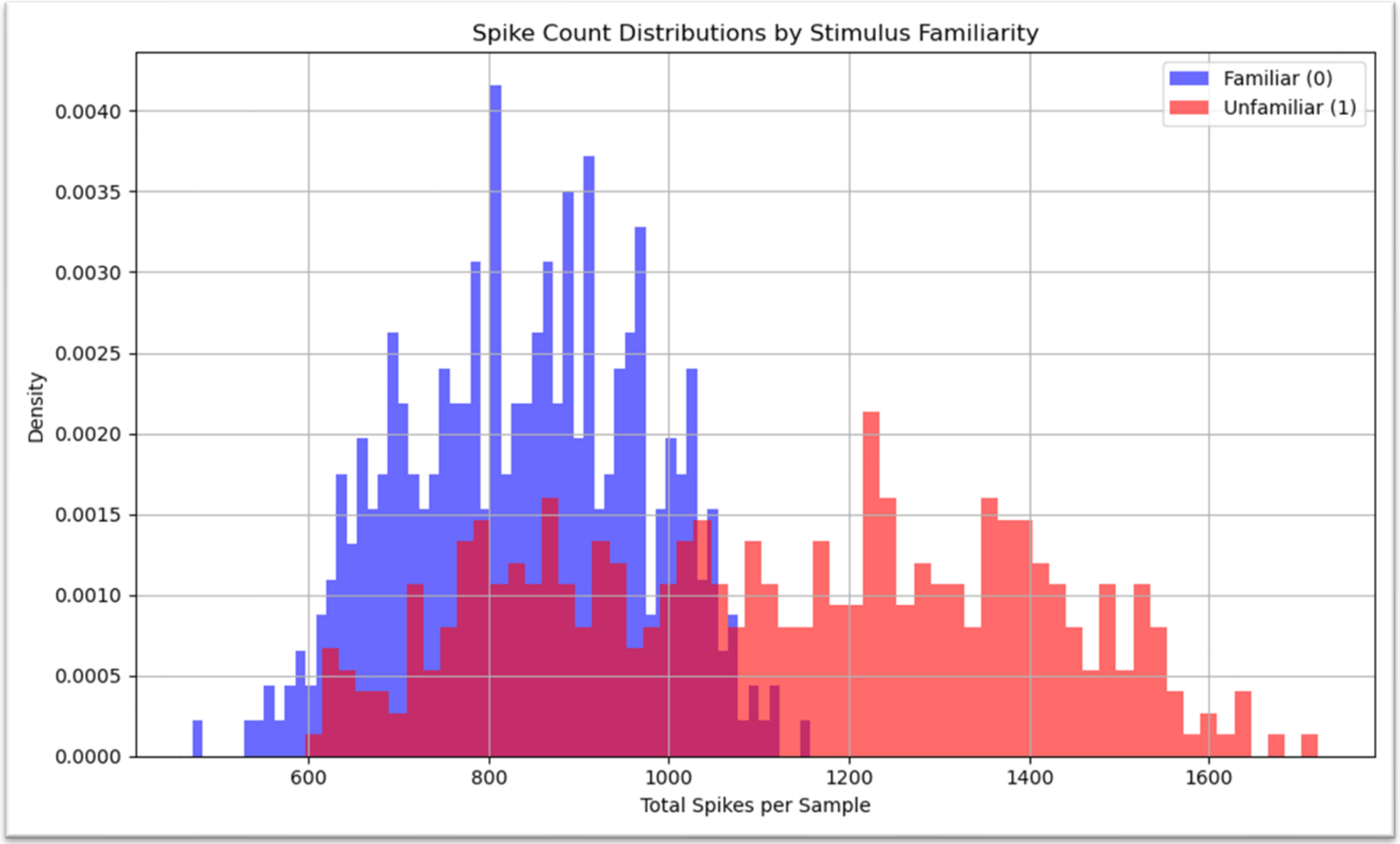
- Description:
  - information is encoded in the firing rate i.e. the number of spikes over time
  - each sample is transformed into ist total number of spikes
- Reasons for choosing this method:
  - based on the second plot from step 1 I derived that the total number of spikes in a sample could be a good feature for a binary classifier
- Transformed dataset:
  - shape: (800, 1)
  - first 10 transformed samples: [471 626 611 698 669 615 633 596 822 777]

Firing Synchrony

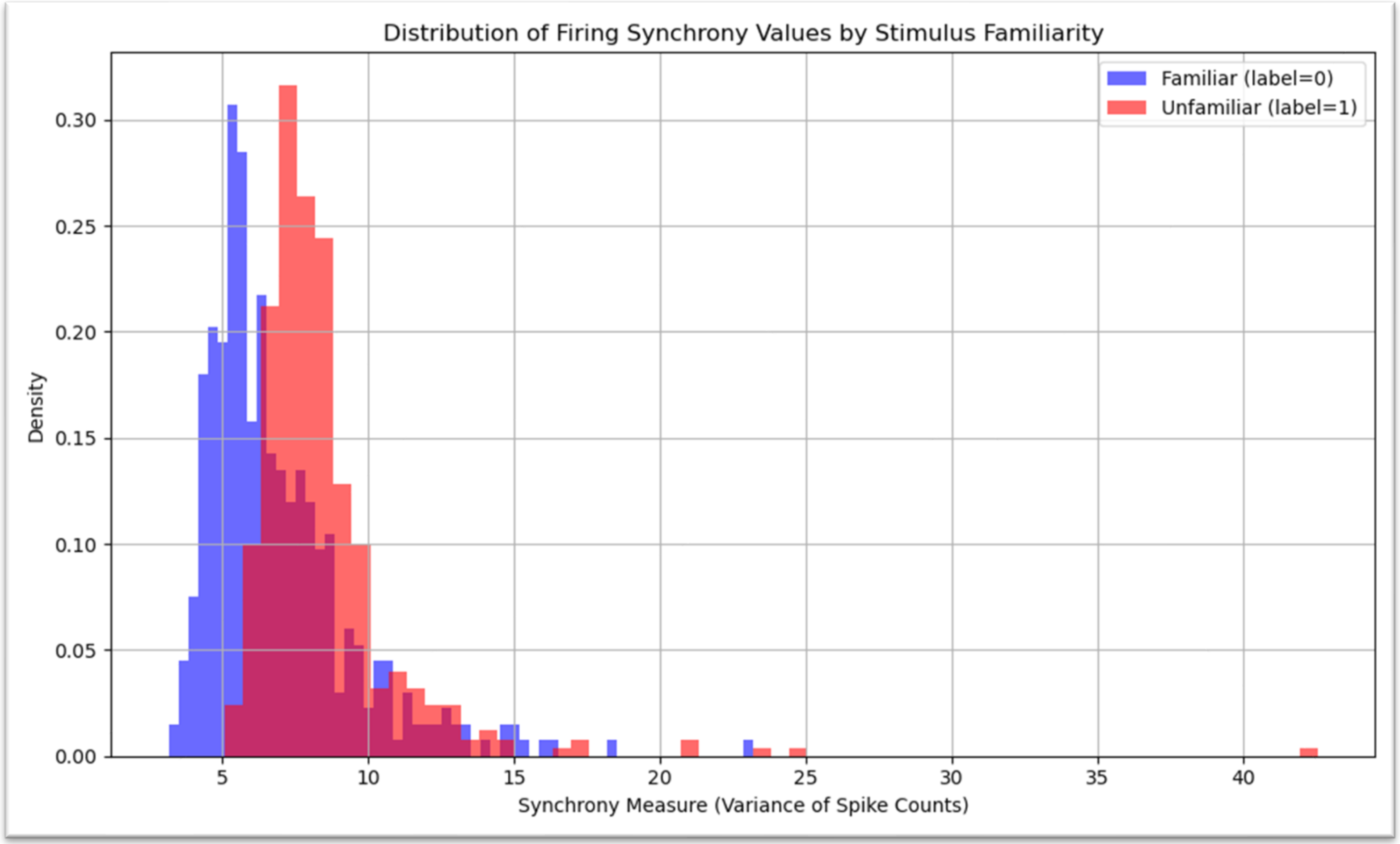
- Description:
  - type of temporal coding where information is encoded by the timing (pattern) of spikes
  - synchrony means “how similarly the neurons fire over time”
  - chosen synchrony measure: variance of spike counts in the population (higher variance = more synchrony)
- Reasons for choosing this method:
  - based on the first plot from step 1
- Transformed dataset:
  - shape: (800, 1)
  - First 10 transformed samples: [8.8067936] [9.10142191] [14.73822017] [14.47013359] [13.34064805] [11.65680615] [16.07030251] [7.88220764] [23.16433002] [20.74241935]

Step 3: Transformed Data – Exploratory Analysis

Rate Coding – Distribution Plot



Firing Synchrony – Distribution Plot



Measureing the difference between the distributions – KL divergence

- Kullback–Leibler divergence
- measure of how one probability distribution diverges from a second, expected probability distribution
    - useful for comparing how one distribution differs from another, which fits the goal of measuring how neural responses change with stimulus familiarity
- higher KL divergence implies greater separability between the two distributions
- computationally inexpensive and easy to implement

KL divergence

Rate Coding Data

KL Divergence (Familiar || Unfamiliar): 1.0852

KL Divergence (Unfamiliar || Familiar): 7.7191

- asymmetry
- familiar spike patterns are somewhat predictable relative to unfamiliar patterns
- unfamiliar stimuli evoke significantly different firing behaviors compared to familiar ones

Firing Synchrony Data

KL Divergence (Familiar || Unfamiliar): 3.0190

KL Divergence (Unfamiliar || Familiar): 0.8189

- familiar stimuli cause a bigger change in firing synchrony compared to unfamiliar stimuli
- In contrast to rate coding, here the familiar patterns are less predictable based on the unfamiliar ones

Step 4:
Model Creation and Training

Chosen Model for firing rate data

- Model 1:
  - Logistic Regression Model
  - both distribution plot and kl divergence support a moderate separability between the two classes (familiar and unfamiliar)
  - linear model should reflect expected direct relationship between spike count and stimulus similarity
  - simple, easy to implement and easy to interpret

Chosen Models for firing synchrony data

- Model 2
  - Logistic Regression Model
  - distribution plot and KL divergence did not indicate same level of separability
  - linear model was still chosen to allow a direct comparision between the two datasets without model complexity being a factor
- Model 3:
  - Random Forest Model
  - try to achieve better results than model 2
  - distributions overlap and are positively skewed
  - Random Forest is a nonlinear model which migth work better and it‘s robust to skewed data

Step 5:
Model Evaluation

Chosen metrics to evaluate the model performance

- same metrics were chosen for both models
- the overall dataset is balanced and has an equal number of samples with a familiar and unfamiliar stimulus
  - split of training and testing data maintains this due to used stratification
- Accuracy
  - indicating how often the model was right
  - due to the balanced character of the data this metric is already meaningful on ist own
- F1 Score
  - added for completness but not strictly necessary in this context
- Confusion Matrix
  - includs true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN)
  - used to understand types of errors

Evaluation Model 1

model correctly classifies most samples, which measn overall firing rate contains meaningful information about stimulus familiarity

F1-scores are similar for both classes, which means the model doesn't favor one

both observations are also supported by the values in the confusion matrix

```
Test Accuracy: 0.7188
Confusion Matrix:
[[61 19]
 [26 54]]

Classification Report:
              precision    recall  f1-score   support

           0       0.70      0.76      0.73        80
           1       0.74      0.68      0.71        80

    accuracy                           0.72       160
   macro avg       0.72      0.72      0.72       160
weighted avg       0.72      0.72      0.72       160
```

Evaluation Model 2

model performes above chance but has a lower accuracy than model 1

F1-scores show slight imbalance with the model performing somewhat better at recognizing familiar stimuli

both observations are also supported by the values in the confusion matrix

```
Test Accuracy: 0.6438
Confusion Matrix:
[[57 23]
 [34 46]]

Classification Report:
              precision    recall  f1-score   support

           0       0.63      0.71      0.67        80
           1       0.67      0.57      0.62        80

    accuracy                           0.64       160
   macro avg       0.65      0.64      0.64       160
weighted avg       0.65      0.64      0.64       160
```

Evaluation Model 3

- model correctly predicts stimulus class in 70% of the test samples
- F1-scores are similar for both classes, which means the model doesn’t favor one
- both observations are also supported by the values in the confusion matrix

```
Test Accuracy: 0.7000
Confusion Matrix:
[[58 22]
 [26 54]]

Classification Report:
              precision    recall  f1-score   support

           0       0.69      0.72      0.71        80
           1       0.71      0.68      0.69        80

    accuracy                           0.70       160
   macro avg       0.70      0.70      0.70       160
weighted avg       0.70      0.70      0.70       160
```

Step 6:
Model Comparision

Comparision Table

- table displays evaluation metrcis for all models side-by-side
- simple way to compare them and identify obvious performance differences

- model 1 outperforms model 2 and 3 across all metrics
- model 3 outperforms model 2 across all metrics
- overall firing rate in a sample is the better predictor for stimulus familiarity

	Metric	Model 1	Model 2	Model 3
0	Accuracy	0.7188	0.6438	0.7000
1	Precision (0)	0.7011	0.6264	0.6905
2	Recall (0)	0.7625	0.7125	0.7250
3	F1-score (0)	0.7305	0.6667	0.7073
4	Precision (1)	0.7397	0.6667	0.7105
5	Recall (1)	0.6750	0.5750	0.6750
6	F1-score (1)	0.7059	0.6174	0.6923